

Machine Learning and Geostatistical Methods for Spatial Prediction of Arsenic in Soils: A Systematic Review (2015–2025)

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Abstract. Machine learning (ML) and geostatistical methods represent a rapidly evolving data-driven paradigm for environmental spatial prediction. This systematic review evaluates and compares the predictive accuracy of geostatistical, ML, and hybrid ML–kriging models for spatial prediction of soil arsenic (As), following the PRISMA 2020 framework across 31 peer-reviewed studies (2015–2025) retrieved from Scopus and SciELO databases. Hybrid ML methods emerged as the dominant approach (55% of studies), consistently outperforming classical geostatistics: the highest validated accuracies were achieved by elastic net regression with diffuse reflectance spectroscopy ($R^2 = 0.97$; RMSE = 0.24 mg kg⁻¹; RPD = 6.32) and GA-PLSR with multi-sensor satellite imagery ($R^2 = 0.88$; RPD = 5.02), while classical ordinary kriging (OK) showed moderate accuracy ($R^2 = 0.45$ –0.72). Auxiliary data sources—portable XRF, hyperspectral imagery, and satellite-derived covariates—were the dominant determinants of predictive performance. A geographic bias was identified (48% of studies from China), with the Andean region of Latin America entirely absent despite high geogenic As burden. Standardized validation protocols and AI-based integration of multi-source environmental data are the primary recommendations of this data science synthesis.

Keywords: Machine learning · Geostatistics · Arsenic spatial prediction · Hybrid models · Spatiotemporal data · Soil contamination · Systematic review · PRISMA 2020

1 Introduction

The spatial prediction of environmental contaminants from heterogeneous, multivariate datasets is a canonical problem in environmental data science and AI-based geospatial analysis. Arsenic (As) is a metalloid of global environmental concern, distributed in soils through weathering of sulfide minerals, hydrothermal processes, and anthropogenic activities including mining, smelting, and pesticide application [24]. Classified as a Group I carcinogen by the International Agency for Research on Cancer [32], chronic As exposure is linked to elevated risks of

lung, bladder, and skin cancers, as well as cardiovascular disease and developmental toxicity [7]. Approximately 140 million people worldwide are exposed to As concentrations above regulatory thresholds, predominantly through contaminated groundwater and food crops grown on As-enriched soils [21].

Accurate spatial prediction of As distribution in soils is foundational to environmental risk assessment, remediation design, and land-use regulation. This structural duality between point-source contamination (steep gradients near mine tailings) and diffuse geogenic inputs challenges conventional interpolation and demands data-driven methods capable of capturing both broad spatial trends and local variability simultaneously.

Geostatistical methods—principally ordinary kriging (OK) and its variants—have been the dominant paradigm for heavy metal spatial prediction since their systematic formalization by [6] and [31]. Kriging provides minimum-variance unbiased linear estimators and generates spatial uncertainty maps, advantages that purely deterministic interpolators such as inverse distance weighting (IDW) cannot offer. Nevertheless, OK assumes spatial stationarity and relies solely on the spatial covariance of the target variable, limiting its performance when As concentrations are governed by complex, non-linear geochemical and anthropogenic interactions.

The past decade has witnessed a data science transition driven by two convergent developments: (i) the proliferation of high-resolution auxiliary data sources—portable X-ray fluorescence (PXRF) spectrometers, satellite hyperspectral imagery, and digital elevation models—and (ii) the maturation of machine learning (ML) and AI algorithms capable of modelling complex non-linear relationships between As and environmental covariates. Hybrid approaches including Random Forest kriging (RFOK), LightGBM-kriging (LGBK), and neural network–kriging hybrids have demonstrated substantial accuracy improvements over standalone OK [25, 26]. Simultaneously, stochastic simulation methods—sequential Gaussian simulation (SGS) and stochastic partial differential equations (SPDE)—provide probabilistic contamination delineation essential for risk-based management [28, 30].

Despite growing methodological sophistication, no systematic data science synthesis of comparative performance specifically for As in soils exists. The present review addresses four research questions: **RQ1:** What geostatistical and ML methods have been applied for spatial prediction of As in soils (2015–2025)? **RQ2:** What is their comparative predictive performance? **RQ3:** What auxiliary variables and contextual factors determine predictive accuracy? **RQ4:** What methodological gaps exist and where are future data-driven approaches most needed?

2 Materials and Methods

This review followed PRISMA 2020 [19] and the PECO framework (Population, Exposure, Comparison, Outcome).

2.1 Eligibility Criteria

Population: Soils with arsenic (geogenic or anthropogenic), no geographic restriction. **Exposure:** At least one geostatistical or ML method for As spatial prediction (OK, co-kriging, SGS, regression kriging, or hybrid ML–geostatistical approach). **Comparison:** Comparison between methods, or standalone evaluation with quantitative cross-validation. **Outcome:** RMSE, MAE, R^2 , RPD, RPIQ, F_1 -score, or equivalent.

Inclusion criteria (all must be met): (I1) original peer-reviewed article; (I2) published 2015–2025; (I3) soil as primary matrix; (I4) As quantified individually; (I5) at least one geostatistical or ML spatial method; (I6) quantitative validation metrics reported; (I7) full text in English, Spanish, or Portuguese.

Exclusion grounds: (E1) not original article; (E2) matrix not soil; (E3) no geostatistical or ML spatial method; (E4) As not individually analyzed; (E5) no validation metrics; (E6) full text inaccessible; (E7–E8) outside language or date range.

2.2 Search Strategy

Systematic searches were conducted on 3 June 2026 in Scopus (primary) and SciELO (regional supplement). The Scopus TITLE-ABS-KEY query was:

(arsenic OR "As contamination" OR metalloid)
AND (soil* OR topsoil OR "mine soil")
AND (geostatistic* OR kriging OR cokriging
OR "spatial prediction" OR variogram
OR "sequential gaussian simulation")

Filters: 2015–2025; Article; English, Spanish, Portuguese.

2.3 Study Selection and Data Extraction

Records were managed in Zotero 9 (deduplication) and screened in Rayyan [18] (blind mode, two stages). A standardized extraction matrix captured: study identification; design (samples, depth, context); As characteristics; geostatistical and ML methods; validation metrics; and principal findings.

2.4 Quality Assessment

Quality criteria adapted from QUADAS-2 [31]: (i) $n \geq 100$ samples; (ii) independent validation; (iii) ≥ 2 complementary metrics; (iv) spatial representativeness of sampling design.

3 Results

3.1 Study Selection

Figure 1 presents the PRISMA 2020 flow diagram. The search yielded 290 records (Scopus: 282; SciELO: 8). After deduplication ($n = 2$), 288 records were screened; 220 were excluded at title/abstract stage. Full-text assessment of 68 records led to 37 further exclusions (including one post-screening exclusion of a radon mortality study confirmed as non-soil-arsenic at full-text review). Final corpus: **31 studies**.

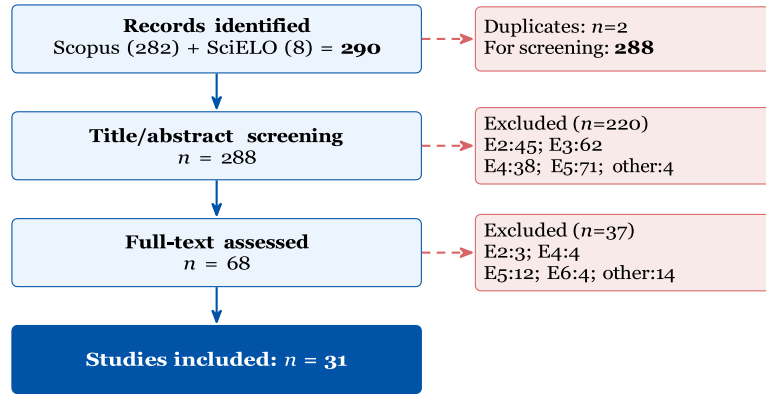


Fig. 1. PRISMA 2020 flow diagram. Exclusion codes defined in Sect. 2.1.

3.2 General Characteristics

The 31 studies were published in 15 journals, with *Science of the Total Environment* ($n = 7$), *Journal of Hazardous Materials* ($n = 5$), *Geoderma* ($n = 4$), and *Environmental Pollution* ($n = 4$) as the most frequent outlets. Publication frequency increased markedly after 2019 (Fig. 2). Studies originated from 10 countries, with China predominating ($n = 15$, 48%), followed by Mexico ($n = 2$, 6%) and the Czech Republic ($n = 2$, 6%). Sample sizes ranged from 25 [23] to 2,029 [10] (median ≈ 150). Study contexts comprised mining-impacted ($n = 15$, 48%), agricultural ($n = 10$, 32%), and urban/industrial ($n = 4$, 13%), with two 3D/subsurface studies (7%).

3.3 Geostatistical and ML Methods Applied

Hybrid ML methods were the most frequently identified as best performer ($n = 17$, 55% of studies), encompassing Random Forest kriging (RFOK), LightGBM-kriging (LGBK), elastic net regression, BP neural networks, and land-use regression (LUR). Classical ordinary kriging (OK) and deterministic methods (IDW,

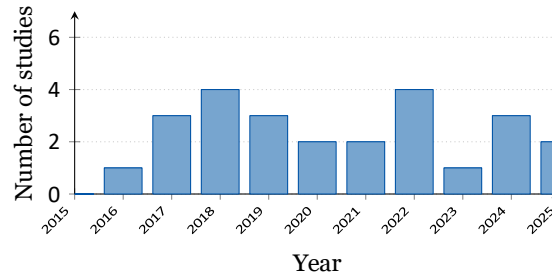


Fig. 2. Annual publication frequency of the 31 included studies (2015–2025).

RBF) remained best performer in 8 studies (26%), stochastic simulation methods (SGS, SPDE) in 4 studies (13%), and co-kriging with auxiliary variables in 2 studies (6%). Table 1 summarizes method frequencies and typical R^2 ranges.

Table 1. Frequency and typical predictive performance of geostatistical and hybrid ML methods ($n = 31$ studies).

Method	n	R^2 (As)
Hybrid ML (RFOK, LGBK, etc.)	9	0.57–0.97
Ordinary Kriging (OK)	8	0.45–0.66
SGS / SPDE (stochastic sim.)	4	F_1 : 59–66%
Regression Kriging (RK)	3	0.52–0.55
Co-Kriging + auxiliary	3	0.48–0.88
IDW / RBF (deterministic)	4	Rarely rep.

ML: machine learning; rep.: reported; SGS: sequential Gaussian simulation.

3.4 Comparative Predictive Performance

Table 2 presents best-validated As prediction performance for the 12 studies that reported complete quantitative metrics. Spectroscopic ML methods achieved the highest R^2 values (0.79–0.97), followed by hybrid RF–kriging (0.57–0.88), co-kriging with auxiliary data (0.48–0.79), and classical OK (0.45–0.66).

3.5 Role of Auxiliary Variables

The type and quality of auxiliary variables was the dominant factor determining predictive performance across all method categories (answering RQ3).

Table 2. Best-validated predictive performance for arsenic across studies reporting complete quantitative metrics. N/R: not reported; DRS: diffuse reflectance spectroscopy; ATPRK: area-to-point regression kriging; FKM: fuzzy k-means.

Study	Best method	R^2	RMSE	RPD	Context
[1]	Elastic Net + DRS	0.97	0.24 mg kg ⁻¹	6.32	Geogenic; India
[13]	GA-PLSR + ATPRK	0.88	9.42 mg kg ⁻¹	5.02	Mining; Iran
[12]	GA-PLSR (VIS-NIR)	0.79	0.149 mg kg ⁻¹	2.33	Mining; Iran
[4]	Random Forest	0.66	N/R	N/R	Agricultural; China
[25]	RF + OK (RFOK)	0.66	N/R	N/R	Oil mining; China
[26]	LightGBM + Kriging	0.63	N/R	N/R	Coal mining; China
[11]	Lasso/ENET + OK	0.62	0.11 mg kg ⁻¹	N/R	Mining; Czech Rep.
[10]	RF + FKM zones	0.57	4.84 mg kg ⁻¹	N/R	Agricultural; China
[20]	Regression Kriging	0.55	N/R	N/R	Smelter 3D; Serbia
[33]	COK + PXRF	0.48	4.64 mg kg ⁻¹	N/R	Agr./mining; China
[2]	Co-Kriging (spectra)	N/R	0.559 mg kg ⁻¹	N/R	Agricultural; China
[17]	RBF / OK	N/R	0.76–0.80	N/R	Mining; Mexico

Portable XRF (PXRF). Used in three studies [8, 14, 33]. PXRF-augmented co-kriging improved R^2 for As by 7–30% relative to OK alone. The 90-second per-sample acquisition time enables much denser spatial coverage than laboratory ICP-AES, directly improving semivariogram estimation and kriging precision.

Hyperspectral and VIS-NIR Spectra. The mechanistic basis lies in the strong adsorption of As onto Fe-oxyhydroxide surfaces, which produce characteristic VIS-NIR absorption features. OSC-transformed spectra achieved $r = 0.898$ in co-kriging [2]; elastic net-selected features achieved $R^2 = 0.97$ with RPD = 6.32 [1], the highest accuracy recorded in this review.

Satellite Imagery. Sentinel-2 and Landsat-8 imagery fused via ATPRK, then fed to GA-PLSR, achieved $R^2 = 0.88$ for As [13]. Satellite covariates are particularly valuable for large-area mapping where dense sampling is logistically impractical.

Environmental and Topographic Variables. Spatial regionalization via fuzzy k-means (FKM) before fitting zone-specific RF models improved As prediction from $R^2 = 0.14$ (global PLSR) to $R^2 = 0.57$ [10], demonstrating the critical importance of managing spatial non-stationarity in AI-based spatial prediction.

3.6 Stochastic Simulation

Four studies employed stochastic simulation rather than deterministic kriging. SGS combined with adaptive sampling improved As delineation F_1 -score from 59 to 66% [28]. SPDE implemented via INLA outperformed SGS in both accuracy and computational efficiency [30]. [15] integrated SGS with Info-Gap Decision Theory (IGDT) for robust multi-metal delineation under severe uncertainty.

3.7 Methodological Quality

Among 31 studies: 39% reported R^2 ; 39% reported RMSE; only 16% reported three or more complementary metrics simultaneously. Validation protocols: k -fold or LOO (52%), holdout (29%), unclear (19%). Spatially-blocked cross-validation [22] was applied in only 10% of studies. Table 3 summarizes quality dimensions across the 31 included studies.

Table 3. Methodological quality assessment of the 31 included studies across four quality dimensions.

Quality criterion	Met	%
$n \geq 100$ samples	18	58%
Independent validation set	22	71%
≥ 2 complementary metrics	12	39%
Spatial validation (blocked CV)	3	10%
R^2 reported	12	39%
RMSE reported	12	39%
Auxiliary variables used	19	61%

4 Discussion

4.1 Performance Hierarchy (RQ1, RQ2)

Our synthesis establishes a clear data-driven hierarchy: hybrid ML–geostatistical methods with auxiliary variables > co-kriging with auxiliary data > regression kriging > classical OK. This aligns with theory: As distribution is a multivariate, non-linear function of soil geochemistry, parent material, and anthropogenic loading—factors that ML algorithms are inherently better equipped to model than the univariate covariance structure assumed by OK. The advantage of hybrid methods was most pronounced in mining contexts, where non-stationarity is extreme and OK’s stationarity assumption is most severely violated.

The persistent use of OK (26% of studies as best performer) reflects its methodological accessibility and its role as a benchmark, rather than superior performance. Studies directly comparing OK with hybrid approaches reported R^2 improvements of 15–100% in favor of hybrids [3, 25, 26].

4.2 Machine Learning as an Emerging Trend (RQ2)

A key finding of this review is the rapid emergence of AI-based and hybrid ML methods as the dominant paradigm for As spatial prediction. Random Forest (RF) and its kriging hybrids (RFOK) consistently appeared as top performers across agricultural and mining contexts [4, 10, 16, 25]. LightGBM-kriging (LGBK), a more recent gradient-boosted hybrid, achieved $R^2 = 0.63$ for sparse mining datasets [26], demonstrating the capacity of ensemble tree methods to handle imbalanced spatial data. Neural network hybrids (BP neural network + kriging) achieved the lowest RMSE in 3D subsurface contexts [9, 27], where geostatistics alone cannot capture non-linear vertical gradients.

The integration of ML with domain-specific geostatistical residual correction (regression kriging, co-kriging) follows a general pattern in data science: combining the interpretability and spatial structure-awareness of kriging with the non-linear feature-learning power of ML. This hybrid architecture has now been validated across mining [25], agricultural [10], and industrial [3] contexts for As prediction, suggesting it represents a robust, transferable framework for environmental contamination mapping.

4.3 Spectroscopic ML Superiority: Mechanistic Basis

The two highest R^2 values (0.97 and 0.88) were both achieved by spectroscopic ML approaches [1, 13]. As in soils is strongly adsorbed onto Fe-oxyhydroxide surfaces and clay mineral edges, which produce characteristic absorption features in the VIS-NIR spectrum (400–2500 nm). The principal limitation is sensitivity to soil moisture and the requirement for a representative calibration dataset for ML models.

4.4 Spatial Non-stationarity and Regionalization (RQ3)

[10] demonstrated a fourfold R^2 improvement (0.14 to 0.57) by decomposing the study domain into environmental subzones via fuzzy k-means before fitting zone-specific RF models. Similar benefits were reported by [5] (BME-GWR) and [20] (layer-specific 3D RK). These findings indicate that the appropriate AI modeling unit for contaminated landscapes is geochemically defined subregions—not administrative boundaries—within which stationarity is locally satisfied.

4.5 Limitations of the Evidence Base (RQ4)

Three limitations constrain the synthesis. First, *metric heterogeneity*: only 39% of studies reported R^2 ; only 16% reported three complementary metrics simultaneously, preventing full cross-study comparison. Second, *validation inconsistency*: spatially-blocked cross-validation [22] was applied in only 10% of studies, biasing R^2 upward due to spatial autocorrelation in standard k -fold splits. Third, *geographic concentration*: 48% of studies from China limits transferability of trained ML models to other geological and pedological contexts.

4.6 Implications for the Andean Region (RQ4)

The Andean region of South America possesses some of the world's highest As concentrations in soils, yet is entirely absent from the systematic ML and geostatistical literature. The Circum-Lake Titicaca metallogenic belt (Puno, Peru) contains geogenic As up to 808 mg kg⁻¹—40 times the WHO threshold [29]. Hybrid RF-kriging incorporating Andean-relevant covariates (volcanic parent material, altitude, proximity to mining districts, irrigation networks) represents a high-priority AI-based methodological adaptation that could substantially advance data-driven contamination mapping in Latin America.

4.7 Recommendations

1. Report RMSE, MAE, and R^2 simultaneously for all ML and geostatistical models.
2. Apply spatially-blocked cross-validation [22] to avoid optimistic bias in AI model evaluation.
3. Integrate PXRF and satellite imagery as standard auxiliary variable practice in hybrid ML-kriging pipelines.
4. Explicitly report sample size relative to the $n \geq 100$ threshold [31].
5. Prioritize understudied high-burden regions for future ML model development: Andean Latin America, sub-Saharan Africa, South/Southeast Asia.

4.8 Limitations of This Review

Restriction to three languages may have excluded relevant Chinese-language ML literature. Single-reviewer screening introduces potential selection bias. Incomplete metric extraction for studies with inaccessible full texts may introduce reporting bias. Formal meta-analytic pooling was not feasible given high between-study heterogeneity in ML model architectures and validation protocols.

5 Conclusions

This systematic review of 31 studies (2015–2025) establishes six evidence-based conclusions answering RQ1–RQ4:

1. Hybrid ML-geostatistical methods (RFOK, GA-PLSR, LGBK, elastic net) were the best-performing approach in 55% of studies, with R^2 improvements of 15–100% over classical OK (RQ2).
2. Auxiliary variables (PXRF, VIS-NIR spectra, satellite imagery) are the dominant determinants of predictive accuracy; their AI-driven integration is the most effective strategy for improving As spatial prediction (RQ3).
3. Classical OK shows moderate accuracy ($R^2 = 0.45$ – 0.66) and systematically underestimates extreme concentrations due to the smoothing effect, confirming the need for ML-based alternatives.

4. SGS-based adaptive sampling improves As delineation F_1 -score by 7%; SPDE outperforms SGS in accuracy and computational efficiency (RQ2).
5. The Andean region, sub-Saharan Africa, and South Asia are critically under-represented despite high As burden, representing priority areas for future ML model training and deployment (RQ4).
6. Reporting RMSE, MAE, and R^2 simultaneously, and applying spatially-blocked validation, is urgently needed to enable rigorous cross-study meta-analysis of AI-based As prediction models.

Outlook. The integration of machine learning and geostatistical methods represents a highly promising and rapidly consolidating line for data-driven spatial prediction of environmental contaminants. Random Forest, gradient-boosted trees, and neural network hybrids have demonstrated consistent superiority over classical kriging in accuracy, and their combination with high-resolution satellite and spectroscopic data opens new frontiers for large-scale, low-cost contamination monitoring. Future work should prioritize spatially-aware ML architectures, open benchmark datasets, and geographically diverse validation to advance the field toward operational AI-based environmental monitoring systems.

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